

Riemann Sums: Approximating Area Under a Curve

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Name: _____

Date: _____

Learning Objectives

- Approximate the area under a curve using Left Riemann Sums (L_n), Right Riemann Sums (R_n), and Midpoint Riemann Sums (M_n)
- Calculate the width of each subinterval and evaluate the function at the correct sample points
- Compare overestimates and underestimates produced by different Riemann Sum methods for increasing and decreasing functions

Problems

1. Find the width of each subinterval (Δx) when approximating the area under a curve on the interval $[0, 4]$ using 4 equal subintervals.

$$\Delta x = \frac{b - a}{n}$$

2. Using the Left Riemann Sum with 4 rectangles (L_4), list the four left-endpoint x -values on the interval $[0, 4]$ where the heights of the rectangles are evaluated.

$$f(x) = x^2 + 1, \quad [0, 4], \quad n = 4$$

3. Evaluate the function at each of the four left endpoints and find the height of each rectangle for the Left Riemann Sum approximation.

x	$f(x) = x^2 + 1$
0	
1	
2	
3	



4. Using the heights found in Problem 3 and a width of 1 for each rectangle, calculate the Left Riemann Sum L_4 to approximate the area under the curve on $[0, 4]$.

$$f(x) = x^2 + 1, \quad L_4 = \Delta x[f(x_0) + f(x_1) + f(x_2) + f(x_3)]$$

5. Using the Right Riemann Sum with 4 rectangles (R_4), identify the four right-endpoint x -values on $[0, 4]$, evaluate $f(x)$ at each, and compute R_4 .

$$f(x) = x^2 + 1, \quad R_4 = \Delta x[f(x_1) + f(x_2) + f(x_3) + f(x_4)]$$

6. Compare L_4 and R_4 for $f(x) = x^2 + 1$ on $[0, 4]$. State which is an overestimate and which is an underestimate, and explain why in terms of the shape of the function.

$$f(x) = x^2 + 1 \text{ is increasing on } [0, 4]$$

7. Using the Midpoint Riemann Sum with 4 rectangles (M_4) on $[0, 4]$, identify the four midpoint x -values, evaluate $f(x)$ at each, and compute M_4 .

8. Approximate the area under the curve on the interval $[1, 3]$ using the Left Riemann Sum with 4 subintervals.

$$f(x) = x^2 + 2x, \quad [1, 3], \quad n = 4$$

9. Approximate the area under the curve on $[0, 3]$ using the Right Riemann Sum with 6 equal subintervals.

$$f(x) = \sqrt{x + 1}, \quad [0, 3], \quad n = 6$$

10. The exact area under $f(x) = x^2 + 1$ on $[0, 4]$ found using a definite integral is $76/3 \approx 25.33$ square units. Using your results from L_4 , R_4 , and M_4 , calculate the absolute error for each method and determine which Riemann Sum gives the best approximation.

Method	Approximation	Exact Value	Absolute Error
L4	18	25.33	
R4	34	25.33	



Method	Approximation	Exact Value	Absolute Error
M4	25	25.33	



Riemann Sums: Approximating Area Under a Curve — Answer Key

Calculus Worksheet · Grade 11–12 / College Intro Calculus

Answer Key

1. Answer: $\Delta x = 1$

- Identify: $a = 0$, $b = 4$, $n = 4$
- $\Delta x = (4 - 0) / 4 = 4 / 4 = 1$

2. Answer: $x = 0, 1, 2, 3$

- $\Delta x = (4 - 0) / 4 = 1$
- Left endpoints start at $a = 0$ and increase by Δx : $x_{\blacksquare} = 0$, $x_{\blacksquare} = 1$, $x_{\blacksquare} = 2$, $x_{\blacksquare} = 3$

3. Answer: $f(0)=1$, $f(1)=2$, $f(2)=5$, $f(3)=10$

x	$f(x) = x^2 + 1$
0	1
1	2
2	5
3	10

- $f(0) = 0^2 + 1 = 1$
- $f(1) = 1^2 + 1 = 2$
- $f(2) = 2^2 + 1 = 5$
- $f(3) = 3^2 + 1 = 10$

4. Answer: $L_4 = 18$ square units

- $\Delta x = 1$, heights: $f(0)=1$, $f(1)=2$, $f(2)=5$, $f(3)=10$
- $L_4 = 1 \times (1 + 2 + 5 + 10) = 1 \times 18 = 18$

5. Answer: $R_4 = 34$ square units

- $\Delta x = 1$, right endpoints: $x = 1, 2, 3, 4$
- $f(1)=2$, $f(2)=5$, $f(3)=10$, $f(4)=17$
- $R_4 = 1 \times (2 + 5 + 10 + 17) = 34$

6. Answer: $L_4 = 18$ (underestimate); $R_4 = 34$ (overestimate); because f is increasing.

- $f(x) = x^2 + 1$ is strictly increasing on $[0, 4]$
- For an increasing function, left endpoints give rectangle heights below the curve $\rightarrow$ underestimate: $L_4 = 18$
- Right endpoints give rectangle heights above the curve $\rightarrow$ overestimate: $R_4 = 34$



- $\Delta x = 1$; midpoints of $[0,1],[1,2],[2,3],[3,4]$ are $x = 0.5, 1.5, 2.5, 3.5$
- $f(0.5) = 0.25+1 = 1.25$; $f(1.5) = 2.25+1 = 3.25$; $f(2.5) = 6.25+1 = 7.25$; $f(3.5) = 12.25+1 = 13.25$
- $M4 = 1 \times (1.25 + 3.25 + 7.25 + 13.25) = 25$

8. Answer: L4 = 14.75 square units

- $\Delta x = (3-1)/4 = 0.5$; left endpoints: $x = 1, 1.5, 2, 2.5$
- $f(1) = 1+2 = 3$; $f(1.5) = 2.25+3 = 5.25$; $f(2) = 4+4 = 8$; $f(2.5) = 6.25+5 = 11.25$
- $L4 = 0.5 \times (3 + 5.25 + 8 + 11.25) = 0.5 \times 27.5 = 13.75$
- Note: $L4 = 13.75$ square units

9. Answer: R6 $\approx$ 4.74 square units

- $\Delta x = (3-0)/6 = 0.5$; right endpoints: $x = 0.5, 1, 1.5, 2, 2.5, 3$
- $f(0.5)=\sqrt{1.5}\approx 1.225$; $f(1)=\sqrt{2}\approx 1.414$; $f(1.5)=\sqrt{2.5}\approx 1.581$; $f(2)=\sqrt{3}\approx 1.732$; $f(2.5)=\sqrt{3.5}\approx 1.871$; $f(3)=\sqrt{4}=2$
- Sum $\approx 1.225+1.414+1.581+1.732+1.871+2 = 9.823$
- $R6 = 0.5 \times 9.823 \approx 4.91$ square units

10. Answer: Errors: L4 $\approx$ 7.33, R4 $\approx$ 8.67, M4 $\approx$ 0.33; M4 is the best approximation.

Method	Approximation	Exact Value	Absolute Error
L4	18	25.33	7.33
R4	34	25.33	8.67
M4	25	25.33	0.33

- Exact area = $76/3 \approx 25.33$
- $|L4 - \text{Exact}| = |18 - 25.33| = 7.33$
- $|R4 - \text{Exact}| = |34 - 25.33| = 8.67$
- $|M4 - \text{Exact}| = |25 - 25.33| = 0.33$
- M4 has the smallest absolute error (0.33), confirming it is the most accurate of the three Riemann Sum methods.

